Bayesian Network Model for Diagnosis of Social Anxiety Disorder

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***Abstract*— Nowadays a lot of systems are developed to predict or suggest a diagnosis about the health level of a patient for helping physicians in their decisional process. Recent researches prove that decisional systems implemented by Bayesian networks represent an efficient tool for medical healthcare practitioners. Bayesian Networks (BNs) are graphical models with significant capabilities that can be used for medical predictions and diagnosis. Social Anxiety Disorder (SAD) is the third most common psychiatric disorder in America behind depression and alcohol abuse. This paper focuses on the use of Bayesian network in assisting SAD diagnosis, in which SAD is analyzed and modeled by Bayesian networks in two phases: construction of BN structure and Conditional Probability Tables (CPTs). This research provides a Bayesian network-based analysis of data, gathered from a number of engineering students.**

Keywords- Bayesian networks; Social phobia; disease diagnosis; decision making.

I. INTRODUCTION

Social Anxiety Disorder (SAD), also known as Social Phobia, is the most common anxiety disorder, with a lifetime prevalence of about 12.1% [1]. It is common for sufferers of social phobia to self-medicate, especially if they are undiagnosed, untreated, or both; this can lead to alcoholism, eating disorders or other kinds of substance abuse. An early diagnosis may help minimize the symptoms and prevent the development of additional problems, such as depression. Unlike some other psychological problems, SAD is not well understood by the general public or by medical and mental health care professionals. People with social anxiety are misdiagnosed almost 90% of the time.

Nowadays physicians can use a lot of systems and mathematical methods that can be applied to suggest a diagnosis or to give a prognosis about the health level of a patient [2, 3]. In this way, Bayesian networks have become increasingly popular for handling the uncertain knowledge involved in establishing diagnoses of disease and are also used in clinical epidemiology for the construction of disease models [4]. In this paper, Bayesian networks are involved in a psychiatric decision making process for predicting the probability of a patient to suffer from SAD based on detected causes and symptoms.

II. BAYESIAN NETWORKS IN MEDICAL DIAGNOSIS

Bayesian networks were introduced in the 1980's as formalism for representing and reasoning with problems involving uncertainty, adopting probability theory as a basic framework and they have proved to be useful in practical applications, such as medical diagnosis and diagnostic systems [5, 6]. Bayesian networks are Directed Acyclic Graphs (DAG) modeling probabilistic dependencies and independencies among variables. The graphical part of them reflects the structure of a problem while local interactions among neighboring variables are quantified by conditional probability distributions. The nodes and arcs of a BN model represent, respectively, the random variables and the dependencies among the variables. The direction of the arcs represents the relations of consequence-cause among the variables. In a BNs model, a parent node is a cause of a child node [7].

The main efficiency of medical Bayesian networks remains in the diagnosis field which is discussed in the following sections.

III. THE BAYESIAN NETWORK MODELING OF SOCIAL ANXIETY DISORDER

An anxiety disorder is a serious mental illness. SAD is an anxiety disorder with varying degrees of severity. Standardized rating scales such as Social Phobia Inventory (SPIN) can be used for screening SAD and measuring its severity [8].

Whereas a Bayesian network has two parts, the process of BN construction involves constructing structure and conditional probability tables. There are three methods to construct a Bayesian network: manual construction, learning and a combination of them [9]. In this work, we construct the structure of our BN model manually using domain knowledge. In this process, variables and relationships between them should be determined. The first stage in manual construction is the identification of the important variables generally based on interviews with experts and descriptions of the domain. . It is important to limit variables and choose important variables which are target variables and observation (evidence) variables. Target variables are outputs of network and what we want to know and observation variables are inputs of network [9]. After that,

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the dependence and independence relationships among the variables have to be analyzed and expressed in the graphical structure [4].

The symptoms of SAD reflect a fear of being embarrassed or humiliated in front of others. People with social phobia and other highly socially anxious people perceive their own performance in social situations more poorly (X_{14} node), even after differences in actual performance are taken into account [10]. The physical symptoms reported by many persons with social phobia include sweating, blushing, twitching, trembling, palpitations, dry mouth, shaky voice and nausea (X_6-X_{13} nodes, respectively) [11]. Also one of the emotional symptoms of social phobia is excessive self-consciousness and anxiety in everyday social situations (X_{15} node). The arc direction is from disease node (SAD) to symptom nodes. Also, "age", "gender", "no of first degree relatives suffering from anxiety disorders" and "overprotective/overcontrolling parenting" variables were considered as X_1-X_4 nodes. The arc direction is from X_1-X_4 nodes to SAD node (X_5).

Subsequently, Netica software from Norsys [12] is used for the construction of the BN, because of its simplicity and high performance. Netica allows parameter learning from data in order to determine the conditional probability table at each node. In order to learn the CPTs, data were gathered from 140 engineering students. This data was split into ¾ of the cases for a training set and ¼ for a test set. The BN was trained with the training set using Netica.

Netica has provided an interface to test the BN using a case file of test data. The node of interest for prediction (SAD) was treated as "unobserved node". SAD node was considered as a binary node with the two states "Present" and "Absent" for testing. The BN was tested with the test set using Netica. In the test report, the error rate was 8.571%. This means that in 8.571% of the cases, the network predicted the wrong value.

IV. Conclusion and Results

In this article, the Bayesian network modeling of a medical decision making is presented and applied to the diagnosis of social phobia.

As a result, the probability of a certain engineering student to be diagnosed with social phobia, depending on no evidence, is approximately 22.7 %. The influence of changing the values of only one evidence node at a time on the probability of diagnosing a student with social phobia is shown in Fig. 1. With increasing severity of a symptom (except for sweating/X_6 and twitching/X_8), the probability of SAD also increases. As shown in Fig. 1, the highest probability is related to blushing/X_7 severely in social situations. Finally, the model can be an efficient tool for medical healthcare practitioners in diagnosis of anxiety disorder.

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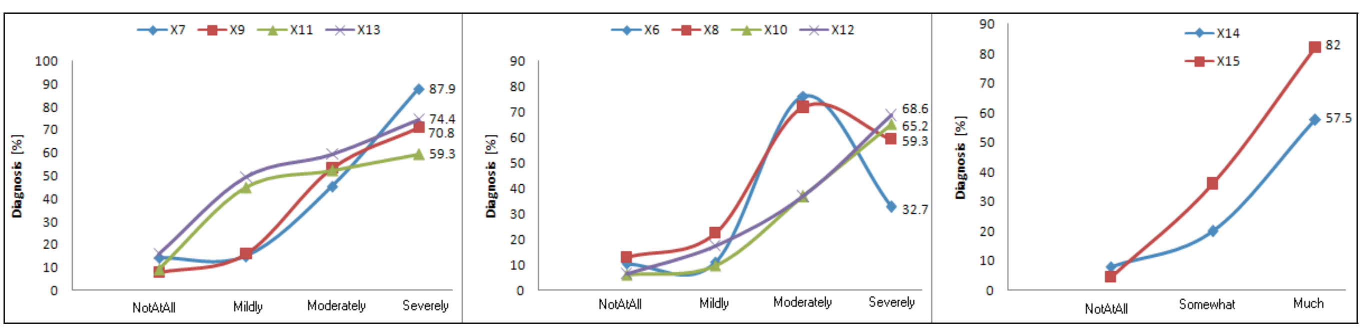

Figure 1. The probability of diagnosing a student with social phobia taking into consideration only one evidence at a time.